

Subject: Computer Graphics

• Section - A (2x5=10)

1. What is Scan Conversion?

Ans: It is a process of representing graphics object as a collection of pixels. The graphics object are continuous. The pixels used are discrete. Each pixel can have either on or off state.

2. What is Rasterization?

Ans: Rasterization is the process of taking a vector graphics image which is made up of shapes and converting it into a raster image which is made up of pixels or dots for output on a video display or for storage in bitmap file format.

3. Define clipping and clip window?

Ans: Clipping: It is the process of removing portions of an image that lie outside a specified viewing area or window.

Clip window: It is the rectangular region in the coordinate system of a graphics device that defines the visible area.

4. Distinguish bet? window port and view port?

Ans: Window port is the world coordinate area to display. In contrast, the view port is the device coordinate area to display.

A window port can be defined with the help of a GWINDOW statement. And a view port is defined by the GPORT command.

5. What is the importance of 8-way symmetry in Scan conversion of circle.

Ans circle is an eight-way symmetric figure. The shape of circle is the same in all quadrants. In each quadrant, there are two octants. If the calculation of the point of one octants is done, the other seven points can be calculated easily by using the concept of eight-way symmetry.

• Section-B (7×3=21)

1. Differentiate between DDA and Bresenham's Line Drawing algorithm.

Ans DDA Line Algorithm	Bresenham's Line algorithm
(i) DDA stands for Digital Differential Analyzer.	(i) No Full Form
(ii) It is less efficient than Bresenham's Line algorithm.	(ii) It is more efficient than DDA algorithm.
(iii) The calculation speed of DDA algo is less than Bresenham's algorithm.	(iii) The calculation is faster than DDA algorithm.
(iv) It is costlier than Bresenham's Line algorithm.	(iv) It is cheaper than DDA algorithm.
(v) DDA algo has less precision or accuracy.	(v) It has more precision or accuracy.
(vi) The complexity of calculation is more complex.	(vi) The complexity of calculation is simple.
(vii) Optimization is not provided.	(vii) Optimization is provided.

3. Explain Bresenham's Line Drawing algorithm. Draw a line from (2,3) to (5,8) by using Bresenham's algorithm.

Ans → Bresenham's Line algorithm:-

The algorithm is used for scan converting a line. It was developed by Bresenham's. It is an efficient

Subject: _____

method because it involves only integer addition, subtraction and multiplication. These operation can be performed very rapidly so line can be generated quickly.

Drawing a line from (2,3) to (5,8) by using Bresenham's algo:

$$x_1 = 2$$

$$x_2 = 5$$

$$\Delta x = 3$$

$$y_1 = 3$$

$$y_2 = 8$$

$$\Delta y = 5$$

$$\text{Slope (m)} = \frac{\Delta y}{\Delta x} = \frac{5}{3} \approx 1.66$$

Here $m \geq 1$

Note, Decision Parameter (P)

$$P = 2\Delta x - \Delta y$$

$$P = 2(3) - 5 = 1$$

$$\therefore P \geq 0$$

if $P \geq 0$, then

$$x_{i+1} = x_i + 1$$

$$y_{i+1} = y_i + 1$$

$$P_{k+1} = P_k + 2(\Delta x - \Delta y)$$

if $P < 0$, then

$$x_{i+1} = x_i$$

$$y_{i+1} = y_i + 1$$

$$P_{k+1} = P_k + 2\Delta x$$

P	x_i	y_i	x_{i+1}	y_{i+1}
1	2	3	3	4
-3	3	4	3	5
3	3	5	4	6
-1	4	6	4	7
5	4	7	5	8

(2,3), (3,4), (3,5), (4,6), (4,7), (5,8)

Subject: _____

Date: __/__/__

3. Write DDA algorithm and also write the advantage and disadvantage of this algorithm.

Ans DDA (Digital Differential Analyzer): - it is a line drawing algorithm used in computer graphics to generate a line segment between two specified endpoints. It is a simple and efficient algorithm that works by using the incremental difference b/w the x-coordinates and y-coordinates of the two endpoints to plot the line.

Steps: -

- (i) input two end points of line (x_1, y_1) & (x_2, y_2)
- (ii) Calculate Δx and Δy .
- (iii) Calculate slope $(m) = \Delta y / \Delta x$
- (iv) Set the initial point as (x_1, y_1)
- (v) Loop through the x-coordinates of the line, incrementing by one each time, and calculate the corresponding y-coordinate using equation $y = y_1 + m(x - x_1)$.
- (vi) Plot the pixel at the calculated (x, y) .
- (vii) Repeat steps 5 and 6 until the end point (x_2, y_2) is reached.

• Advantages:

- it is simple and easy to implement.
- it avoids using multiple operations which have time complexities.

• Disadvantages:-

- it deals with the rounding off operation and floating point arithmetic so it has high time complexity.
- As it is oriented-dependent, so it has poor endpoint accuracy.

Subject: _____

Subject: _____

• sec-C (11x3=33)

2. Write mid point circle generation algorithm. Draw circle having centre (0,0) and radius 9 using this algorithm.

Ans The mid-point circle drawing algorithm is an algorithm used to determine the points needed for rastering a circle.

for any given pixel (x, y),

$$\text{decision parameter } (P) = \frac{5 - r(\text{radius})}{4} \quad \text{or}$$

$$5/4 \approx 1$$

$$P = 1 - r(\text{radius})$$

if $P < 0$ if $P \geq 0$

$$x_{i+1} \rightarrow x_i + 1$$

$$x_{i+1} = x_i + 1$$

$$y_{i+1} \rightarrow y_i$$

$$y_{i+1} = y_i - 1$$

$$P_{i+1} = P_i + 2x_i + 3$$

$$P_{i+1} = P_i + 2(x - y) + 5$$

Given, initial point (0,0)

$$\text{Radius} = 9$$

Initialize $(x_k, y_k) = (0, 9)$

$$P_k = 1 - r = 1 - 9 = -8$$



Octant (I/e)

$$P_k < 0$$

Quadrant 1

$$x_{i+1} = x_i + 1$$

$$(x, y)$$

$$y_{i+1} = y_i$$

$$(0, 9)$$

$$P_{k+1} = P_k + 2(x_i) + 3$$

$$(1, 9)$$

$$(2, 8)$$

$$\text{for } P_k = -8$$

$$(3, 8)$$

$$x_{i+1} \neq 0$$

$$(4, 7)$$

$$y_{i+1} = 9$$

$$(5, 6)$$

$$P_{k+1} = -8 + 2(0) + 3 = -5$$

$$(6, 5)$$

$$\text{for } P_k = -5$$

Subject: _____

Date: ___/___/___

$$y_{i+1} = 9$$

$$p_{k+1} = -5 + (2)(1) + 3 = 0$$

$$\text{for } p_k = 0$$

$$x_{i+1} = 2$$

$$y_{i+1} = 9 - 1 = 8$$

$$p_{k+1} = 0 + 2(2 - 8) + 5 = -7$$

$$\text{for } p_k = -7$$

$$x_{i+1} = 3$$

$$y_{i+1} = 8$$

$$p_{k+1} = -7 + 2(3) + 3 = 2$$

$$\text{for } p_k = 2$$

$$x_{i+1} = 4$$

$$y_{i+1} = 7$$

$$p_{k+1} = 2 + 2(4 - 7) + 5 = 1$$

$$\text{for } p_k = 1$$

$$x_{i+1} = 5$$

$$y_{i+1} = 6$$

$$p_{k+1} = 1 + 2(5 - 6) + 5$$

$$= 1 + (-2) + 5 = 4$$

$$\text{for } p_k = 4$$

$$x_{i+1} = 6$$

$$y_{i+1} = 5$$

$$\text{Hence } x_i \geq y_i$$

Stop

Q2.

Ans

Write Cohen Sutherland algorithm for line clipping.

In this algorithm, we are given 9 region on the screen. out of which one region is at the window and the rest 8 are around it given by 4 digit binary. The division of the regions are based on $(x_{\text{max}}, y_{\text{max}})$ and $(x_{\text{min}}, y_{\text{min}})$. The central part is the viewing region or windows,

Subject: _____

all lines which lie within this region are completely visible.

Algorithm:-

- (i) Calculate position of both endpoints of line.
- (ii) Perform OR operation on both of these end-points
- (iii) If OR operation given 0000
The line is considered visible.

else

perform AND operation on both end points
if AND = 0000

The line is considered the clipped case.

- (iv) If a line is clipped case, find an intersection with boundaries of the window.

$$m = (y_2 - y_1) / (x_2 - x_1)$$

- (a) if bit 1 is "1" line intersects with left boundary of rectangle window

$$y_3 = y_1 + m(x - x_1)$$

where $x = x_{\min}$

where $x_{\min}$ minimum value of x -coordinate

- (b) if bit 2 is "1" line intersects with right boundary

$$y_3 = y_1 + m(x - x_1)$$

where $x = x_{\max}$

- (c) if bit 3 is "1" with bottom boundary

$$x_3 = x_1 + (y - y_1) / m$$

where $y = y_{\min}$

- (d) if bit 4 is "1" line intersects with top boundary

$$x_3 = x_1 + (y_1 - y_1) / m$$

where $y = y_{\max}$.

Subject: _____

Q3. Explain flood filling and boundary filling methods. Give its two real life example where they are used.

Ans

→ Flood Filling :-

Principle : The algorithm starts from a seed pixel and recursively fills adjacent pixels with the same colour until it encounters a boundary of a different color.

Process : (i) Select a seed pixel within the region to be filled.

(ii) check colour of seed pixel.

(iii) Recursively check and fill neighbouring pixels with the same colour if they are within the region.

(iv) Stop when a boundary pixel of different color encountered.

• Boundary Filling :

This algorithm starts from the boundary of the region to be filled moves inward until it reaches the center.

Process : (i) Identify the boundary pixels of the region.

(ii) Start from a boundary pixel and move inward, filling adjacent pixels with same colour until the center is reached.

(iii) Repeat for all boundary pixels.

• Real Life Example :

(i) Color replacement : Both algo can be used to replace a specific color in an image with another color.

(ii) Object filling :- Flood filling can be used to fill object with a solid colour, while boundary filling can be used to fill objects with a gradient or pattern.